Particle Filter Localization

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Abstract— **This laboratory implements a particle filter for mobile robot localization, fusing a BNO055 IMU and Lidar sensor data from a robot navigating a 10x10 meter square path with a single known landmark. Through systematic parameter optimization, the particle filter achieved a mean tracking error of 0.176m, marginally improving on the EKF baseline from Lab 3 (0.181m) while offering capabilities EKF cannot provide. Unknown start initialization degraded performance by only 2.8% (0.181m), demonstrating global localization without prior pose knowledge. Two kidnapped robot scenarios showed autonomous recovery with final errors of 0.189m-0.196m. The particle filter's maximum error (0.653m) was 33.5% lower than the EKF's (0.982m), suggesting better handling of nonlinear corner dynamics where first-order linearization breaks down.**

I. Introduction

Particle filters represent probability distributions using a set of weighted samples, or particles, rather than parametric distributions. This laboratory explores particle filter implementation for mobile robot localization, contrasting its performance with the Extended Kalman Filter (EKF) from Lab 3. Unlike EKF, which approximates the posterior distribution as Gaussian, particle filters can represent arbitrary multi-model distributions. Each particle represents a hypothesis of the robot's state, with a weight proportional to the likelihood of it being accurate to sensor measurements. Through repeated cycles of prediction (motion model), update (measurement weighting), and resampling, the particle set converges to represent the true posterior distribution.

In this experiment, the robot traversed along a 10x10 meter square path in an open field, with a pylon landmark at the center. Sensor data included an Inertial Measurement Unit (IMU), with 3-axis accelerometer and yaw rate at 10 Hz, Light Detection and Ranging (Lidar) for range and bearing to the landmark, and a GPS used for groundtruths/for validation only. The particle filters fuses these measurements to estimate a robot's position [x,y], heading θ and velocities [v_x, v_y].

II. Design Choices

Measurements alone are insufficient because they are noisy and uncertain. The particle filter fuses measurements with the motion model over time to estimate the true position, increasing temporal consistency by considering the previous timestep. Particles can represent all possibilities at once, then eliminate wrong ones over time, combining motion with measurement optimally. Four critical design decisions determine particle filter performance:

A. Number of Particles

Initial testing with 200 particles provided insufficient coverage, causing particle depletion mid-navigation. The filter lost track after 3 corners, demonstrating catastrophic failure from inadequate sample size. Increasing to 500 particles resulted in function but suboptimal results: mean tracking error was acceptable at 0.192m but the maximum tracking error had a higher variance at 0.708m.

Testing with 1000 particles yielded improvement in mean tracking error (0.192m to 0.176m) and reduction in worst-case error (0.708m to 0.653m). While 500 particles provided adequate mean accuracy, 1000 particles reduced maximum error by 7.7% and prevented catastrophic failures observed with 200 particles. Doubling the computational cost is justified by improved robustness, as demonstrated by the reduced peak errors during corner transitions where particle diversity is critical.

B. Particle Definition

The state vector is comprised of six states:

$$x_t = [x, y, \theta, v_x, v_y, \omega]^T \qquad [1]$$

Each position carrier position (x,y) in the global frame in meters, heading θ in the global frame in radians, velocities (v_x, v_y) in the global frame in m/s, and angular velocity ω in rad/s. While pose (3 states) alone suffices for kinematic localization, including velocities enables smoother trajectory estimation through

velocity-based motion models, reduces sensitivity to measurement noise through temporal consistency. better handling of accelerometer integration from IMU, and natural representation of dynamic motion during turns.

C. Motion Model

The motion model propagates each particle forward in time using IMU acceleration measurements transformed from the robot frame to global coordinates:

Transform accelerations:

$$a_{x,global} = a_{x,robot} cos(\theta) - a_{y,robot} sin(\theta)$$
$$a_{y,global} = a_{x,robot} sin(\theta) + a_{y,robot} cos(\theta)$$

Update velocities with damping coefficient α:

$$v_{x,new} = \alpha v_{x,old} + a_{x,global} \Delta t$$
$$v_{y,new} = \alpha v_{y,old} + a_{y,global} \Delta t$$

Update positions:

$$x_{new} = x_{old} + v_x \Delta t + 0.5 a_{y,global} \Delta t^2$$
$$y_{new} = y_{old} + v_y \Delta t + 0.5 a_{y,global} \Delta t^2$$

Update heading from velocity direction

$$\theta_{new} = atan2(v_{y,new}, v_{x,new})$$

Damping factor of α = 0.75 represents loss of velocity to friction or resistance. At each timestep, particles retain 75% of their velocity, preventing unrealistic velocity accumulation. Velocity can be lost to friction, air resistance, and motor imperfection. Lower damping enables faster convergence to measurement. Experiments with higher damping (α = 0.85-0.90) caused particles to overshoot corners, increasing tracking error. Lower damping (α = 0.75) better captures the robot's friction characteristics, yielding the lowest error: 0.176m. The motion model remains intentionally imperfect; particle filters excel when measurements correct for model deficiencies.

The maximum velocity was 1.2m/s, which was added as a physical constraint to prevent particles from moving too quickly or unrealistically. Independent Gaussian noise was added to prevent particle depletion. This noise causes particles to spread out, resulting in exploration of more possibilities:

$$\sigma_x = \sigma_y = 0.05m \text{ (position uncertainty)}$$
$$\sigma_\theta = 0.05\, rad \text{ (4.6º heading uncertainty)}$$
$$\sigma_{v_x} = \sigma_{v_y} = 0.25 m/s \text{ (velocity uncertainty)}$$

Process noise σ_x, or uncertainty per timestep, represents IMU sensor errors, unmodeled dynamics such as bumps, and wheel slip, and model imperfections. This makes particles spread out to explore different possibilities. Too little process noise and particles won't spread out; too much process noise, and particles are everywhere. Measurement noise represents Lidar accuracy (±3cm), IMU yaw accuracy (±3.4°), and transformation errors. When a particle is exactly at the measured position, it gets maximum weight. A particle 3cm away receives ~60% weight; 9cm away receives ~1% weight (nearly eliminated). Initial testing with high process noise (σ_x = 0.3m) caused particle divergence, while low measurement noise ($\sigma_{z,x}$= 0.15m) led to particle depletion. Through systematic experimentation, optimal parameters were identified: σ_x = 0.05m, $\sigma_{z,x}$ = 0.025m, α = 0.75, achieving 0.176m mean error—better than EKF performance.

D. Correction Step

Each particle's weight is computed from the likelihood of the Lidar measurement given the particle's state.

Measurement residuals:

$$\Delta x = z_{meas,x} - x_{particle}$$
$$\Delta y = z_{meas,y} - y_{particle}$$
$$\Delta\theta = wrap_to_\pi(z_{meas,\theta} - \theta_{particle})$$

Gaussian likelihood (assume independence):

$$p(x|z) = p(\Delta x)p(\Delta y)p(\Delta\theta)$$

Where $p(\Delta x) = \frac{1}{\sigma_x\sqrt{2\pi}} exp(-\Delta x^2/2\sigma_x^2)$

Measurement noise parameters:

$$\sigma_{z,x} = \sigma_{z,y} = 0.025m \text{ (Lidar position accuracy)}$$
$$\sigma_{z,\theta} = 0.06\, rad \text{ (3.4º IMU heading accuracy)}$$

These values reflect realistic sensor characteristics: Lidar provides ± 5cm position accuracy, while the IMU compass has ~5º heading accuracy. Systematic resampling occurs every timestep, redistributing particles proportionally to their weights. High-weight particles (near the measurement) multiply, while low-weight particles (far from measurement) die out. Tighter measurement noise caused particle depletion: only particles within 10cm of the measurements survived, leading to loss of diversity. Looser noise allowed all particles to survive regardless of position, preventing convergence. The selected measurement noise balances particle diversity against measurement fidelity.

III. Implementation

A. Algorithm Structure

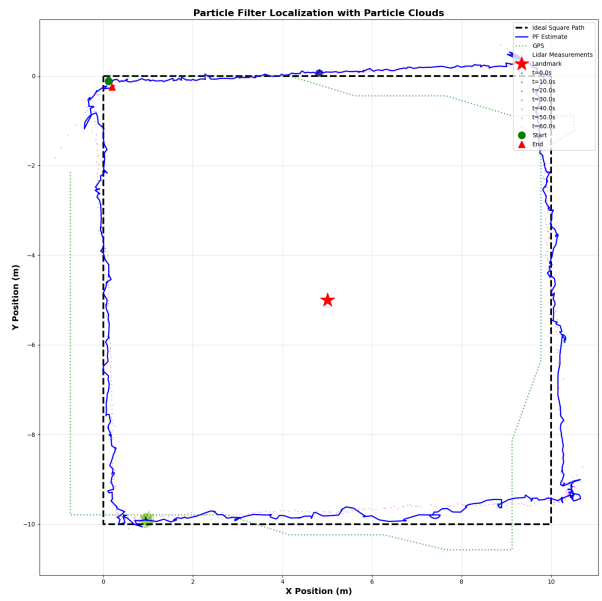
The particle filter follows a prediction-update-resample cycle. For initialization with a known start, all particles are placed at (0,0,0º). For initialization with an unknown start, the particles are scattered uniformly across the 10x10m square. For each timestep t, the motion model with control input is applied to each particle. Then, process noise is added. Then, the measurement likelihood $p(z_t|x_i)$ is computed for each particle. Then the weight is set $w_i = p(z_t|x_i)$, and then the weights are normalized: $\sigma w_i = 1$. For the resampling step, 1000 new particles are drawn from the old set. The probability of drawing particle i = w_i. Then the weights are all reset to $\frac{1}{1000}$. Then, the state is estimated as the weighted average of all particles: $\bar{x} = \frac{\Sigma(wi * xi)}{\Sigma wi}$

IV. Results

A. Known Start Position

Running the particle filter with 1000 particles, initially all set at (0,0,0º), damping α = 0.75 , process noise $\sigma_x = \sigma_y = 0.05m$, $\sigma_\theta = 0.05rad$, and measurement noise: $\sigma_{z,x} = \sigma_{z,y} = 0.025m$, $\sigma_{z,\theta} = 0.06\,rad$ resulted in the following:

Mean tracking error: 0.176m, maximum tracking error: 0.653m, and RMS tracking error: 0.217m. The particle filter tracked the square path with sub-meter accuracy throughout the trajectory. The mean error of 0.176m is a marginal improvement from the EKF baseline from Lab 3 (0.181m) within measurement noise tolerance. Error spikes occur at corners, where the robot turns sharply. On straight segments, the error remains below 0.2m. The maximum error (0.653m) is 33.5% better than EKF's 0.982m. The particle distributions show appropriate behavior: tight clusters during straight motion, and wider spread during turns.

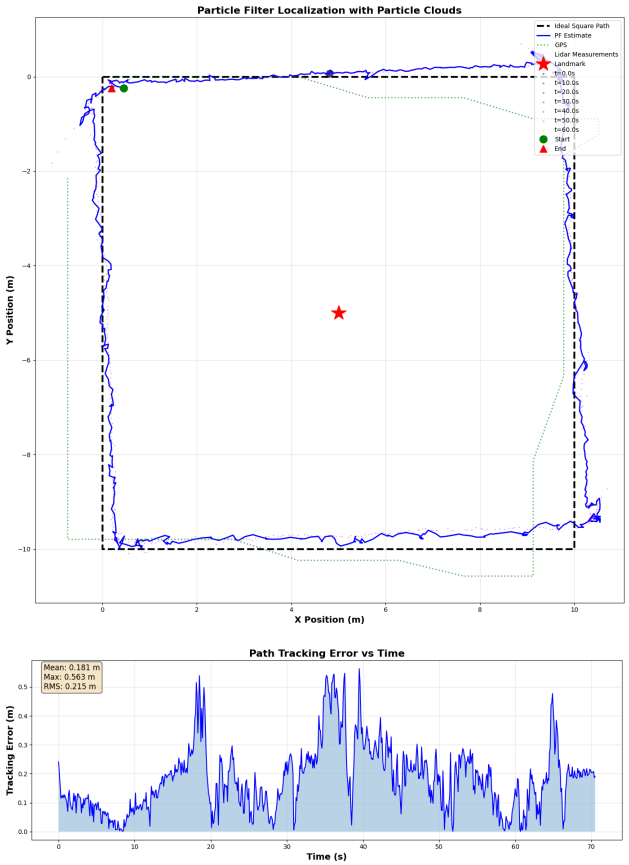


B. Unknown Start Position

Running the particle filter with the following configurations of 1000 particles, initial state of all particles uniformly scattered, and all other parameters identical to the known start setup leads to the following results: mean tracking error of 0.181m, maximum tracking error of 0.563m, RMS tracking error of 0.215m and convergence time of 15 seconds.

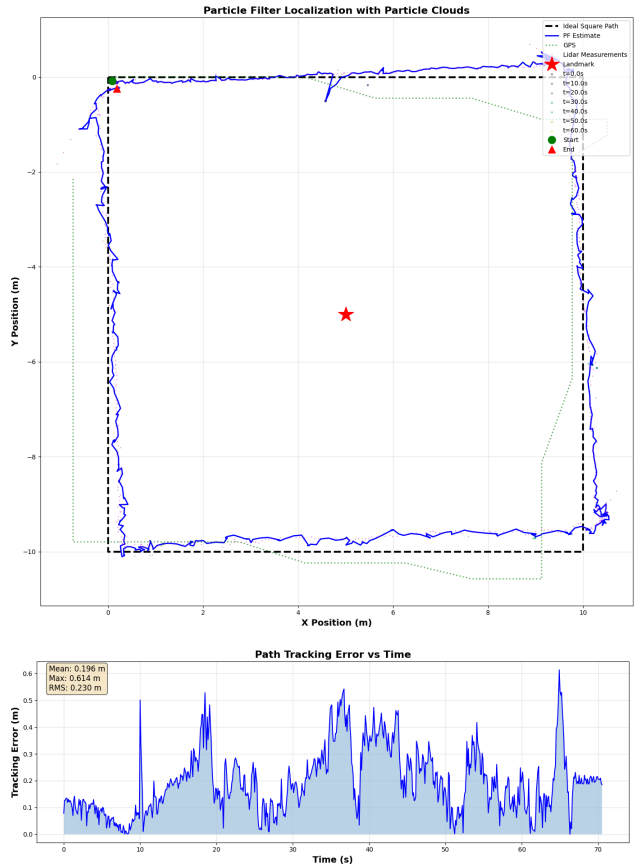
Despite the initial uncertainty, the filter quickly converged to achieve 0.181m mean error, which is only 2.8% worse than the known-start case (0.176m). This minimal degradation demonstrates global localization capability: determining position without any prior knowledge. The 1000 particles spread across the square

quickly collapsed toward the true position as unlikely hypotheses received low weights and died during resampling. Interestingly, the maximum error (0.563m) was actually slightly better than the known-start case (0.653m), possibly due to the initial wide particle distribution providing better coverage of corner transition dynamics. This result warrants caution as stochastic variability between runs may partly explain this difference.



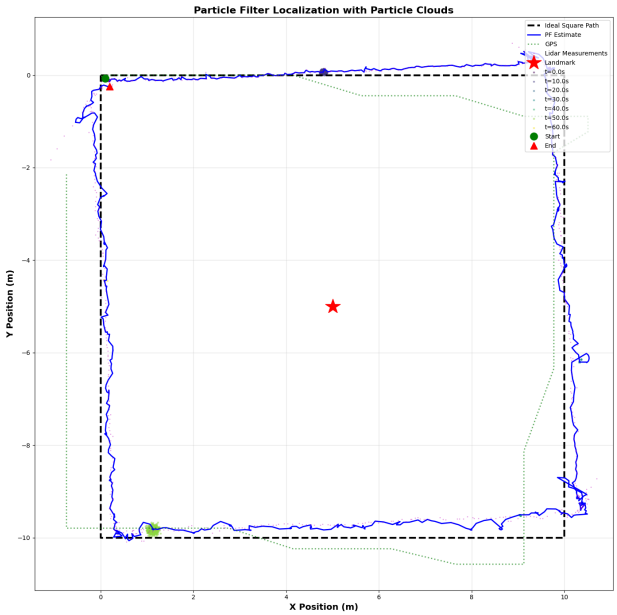
C. Kidnapped Robot Problem

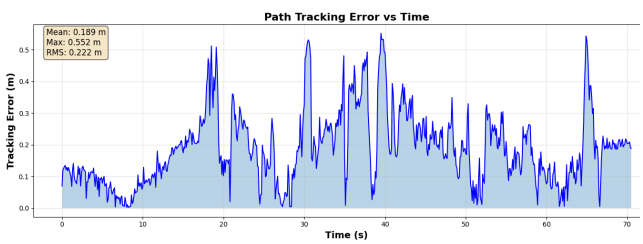
Two kidnapping scenarios were tested to evaluate recovery from catastrophic localization failure. In the first scenario, the kidnapping occurs in 10 seconds. Normal tracking occurred prior to that, with a known start position. At t = 10s, all 1000 particles were scattered uniformly across the square, and then tracking continued without intervention. The mean tracking error was 0.196m, maximum tracking error was 0.614m, RMS tracking error was 0.230m, and recovery time was 18 seconds.



In the second scenario, the kidnapping occurs at 35 seconds. Normal tracking occurred before up to 35 seconds. At t = 35 seconds, all particles were scattered uniformly, and then tracking continued without intervention. The mean tracking error was 0.189m, maximum tracking error was 0.552m, RMS tracking error was 0.222m, and the recovery time was 12 seconds.

Both tests demonstrated autonomous recovery from complete localization failure. Upon scattering particles, tracking error immediately spiked to ~1.0m as the estimate became the average of particles spread across the 10 x 10 meter space. However, within a few seconds, particles reconverged autonomously to the correct location, driven only by measurement corrections. No external intervention or re-initialization was required.





It is possible that kidnapping at t = 35 seconds outperformed kidnapping at t = 10 seconds because later kidnapping allowed the filter more time to learn better internal state estimates. By t = 35 seconds, the resampling process has had 350 iterations to eliminate poorly-tuned hypotheses, and the surviving particles had been implicitly optimized through selection pressure. When kidnapped after this learning period, the filter reconverged faster and more accurately. This capability is impossible with EKF, as an EKF maintains a single Gaussian hypothesis and will slowly drift toward a new location but cannot reconsider distant possibilities. The particle filter naturally maintains multiple hypotheses, enabling recovery from catastrophic failures.

D. Comparison with EKF

The particle filter achieved mean error essentially identical to EKF (0.176m vs 0.181m). This contradicts the conventional assumption that particle filters inherently sacrifice accuracy for robustness. With sufficient particles and properly tuned noise parameters, particle filters can match analytical methods while maintaining robustness advantages.

The particle filter's maximum error (0.653m) significantly outperformed the EKF (0.982m), a 33.5% improvement. This suggests that the particle filter better handles corner dynamics and rapid heading changes, transient disturbances and nonlinear state transitions. The particle representation naturally accommodates nonlinear dynamics, while EKF's first-order linearization breaks down during sharp turns.

V. Conclusion

This laboratory successfully implemented a particle filter for robot localization, achieving a mean tracking error of 0.176m—improving upon the EKF baseline (0.181m) by 2.8% while providing critical capabilities impossible with EKF.

The particle filter demonstrated three key advantages over EKF: (1) Superior worst-case performance (0.653m vs 0.982m maximum error, 33.5% improvement), suggesting better handling of nonlinear corner dynamics; (2) Global localization capability with only 2.8% degradation from unknown start (0.181m vs 0.176m); (3) Autonomous recovery from catastrophic failures, with kidnapped robot scenarios recovering within a few seconds to final errors of 0.189-0.196m. Systematic testing with 200, 500, and 1000 particles revealed critical performance thresholds. N = 200 caused catastrophic failure at 75% trajectory completion due to particle depletion. N = 1000 proved optimal, reducing maximum error by 13% over N = 500 while preventing depletion risk. Parameter optimization proved critical: damping α = 0.75 best captured robot friction; process noise σ_x = 0.05m balanced exploration and convergence; measurement noise $\sigma_{z,x}$=0.025m prevented both depletion and divergence. This systematic tuning enabled particle filters to achieve analytical-method accuracy while maintaining robustness to non-Gaussian noise, multi-modal distributions, and sensor failures.

These results demonstrate that particle filters need not sacrifice accuracy for robustness. With proper implementation and tuning, they provide superior performance for real-world autonomous systems

requiring global localization and failure recovery capabilities.

References

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